Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam B · Form 2: Single-step application

Multiple Choice Questions: Select the best answer. (1 point each)

1. A 30 kg box initially at rest is pushed horizontally by 58.8 N. $\mu_s = 0.4$, $\mu_k = 0.25$. What is the actual friction magnitude? Use g = 9.8 m/s².

(A) 58.8 N

(B) -58.8 N

(C) 117.6 N

(D) 29.4 N

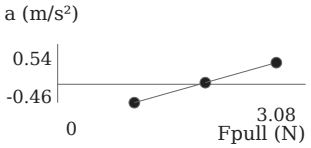
2. A 2 kg cart slides right against constant 2 N friction to the left. Fpull (N) -> a (m/s²): (1.08, -0.46); (2.08, 0.04); (3.08, 0.54). What is the slope between the endpoint measurements?

(A) 0.7

(B) 0.9

(C) 0.3

(D) 0.5



3. A 2 kg cart on a frictionless horizontal plane experiences 12 N east and 7 N north. Find the net force magnitude.

(A) 27.785 N

(B) 6.946 N

(C) 13.892 N

(D) -13.892 N

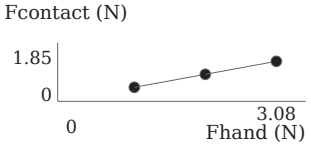
4. Touching 2 kg and 3 kg carts accelerate on a frictionless floor when a hand pushes the 2 kg cart right. Fhand (N) -> Fcontact (N): (1.08, 0.648); (2.08, 1.248); (3.08, 1.848). Which relation models this apparatus?

(A) Y=-0.6X+(0)

(B) $Y=0.6X^2+(0)$

(C) C=m2F/(m1+m2)

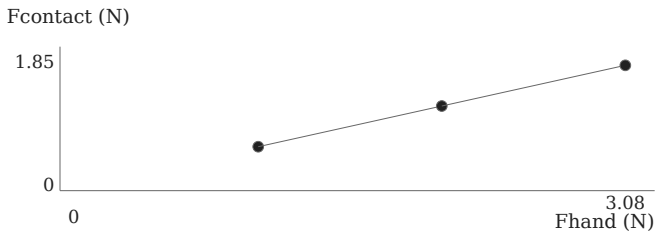
(D) Y=1.2X+(0)



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Touching 2 kg and 3 kg carts accelerate on a frictionless floor when a hand pushes the 2 kg cart right. Fhand (N) -> Fcontact (N): (1.08, 0.648); (2.08, 1.248); (3.08, 1.848).

a. For the two carts in contact setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Label the axes and units and demonstrate interpolation between two readings. **(3 pt)**

6. Release a block on a frictionless ramp and record downhill acceleration. Use g=10 m/s². sinθ -> a (m/s²): (0.1296, 1.296); (0.2496, 2.496); (0.3696, 3.696).

a. Build the model for ramp tilt experiment from physics and predict both slope and intercept. **(2 pt)**

b. Label the axes and units and demonstrate interpolation between two readings. **(2 pt)**

7. A 2 kg cart slides right against constant 2 N friction to the left. Fpull (N) -> a (m/s²): (1.08, -0.46); (2.08, 0.04); (3.08, 0.54). Build the model for cart under tow from physics and predict both slope and intercept. **(3 pt)**